# TCJastrowFitting

December 2025

## To-Do List

- Format Alignment.
  - ✓ Set up the To-Do list.
  - ☐ Set up the format of the raw data.
- Depth-Adjusted ee term investigation.
  - ✓ Alpha interpolation between varmin and emin freely-optimized jastrow (ano-vdz/cc-pvdz/cc-pCVDZ/aug-cc-pvqz)
  - ✓ Freely-optimized varmin AE results with ee cutoff is set to be 6. (aug-ano-v{d/t}z)
  - ✓ Standard freely-optimized emin results of AE (ano-vdz/cc-pCVDZ/aug-ano-v{d/t}z/aug-cc-pvqz)
  - ✓ Standard freely-optimized varmin results of AE (ano-vdz/cc-pCVDZ/aug-ano-v{d/t}z/aug-cc-pvqz).
  - ✓ Ee-depth-fixed AE calculation (ano-vdz, aug-ano-vd/tz; Depth: 0.0, -0.1, -0.2, -0.3, -0.4, -0.5).
  - ✓ AE calculation with the ee depth fixed at the fitting-optimal depth. (aug-ano-vdz: Depth: -0.33,-0.37)
  - ☐ ECP: Fit quadratic functions for total energies and do the same panel plots like for AE.
  - ☐ ECP and AE: Plot difference of total energy of fixed depth and total energy of free floating depth such that we can compare the total energies of AE and ECP in a fair manner.
  - ☐ Plot the free floating e-e depths of AE Jastrows like Kristoffer did for ECPs.
  - ☐ Plot out the max deviation with MAE, RMSE.
- Depth-Adjusted en/een term investigation.
  - ☐ Scale the depth/height of en/een term in Jastrow.

- Spin-dependent Jastrow.
  - ✓ Standard freely-optimized varmin results of AE (Basis: aug-ano-v{d/t}z).
  - ✓ Opposite-spin ee-depth-fixed AE calculation (Basis: aug-ano-vd/tz; Depth: 0.0, -0.1, -0.2, -0.3, -0.4, -0.5).
  - ☐ Ee-cusp spin-dependent Jastrow.

# Results

## 0.1 Free floating depths with varmin optimization (spin-independent)

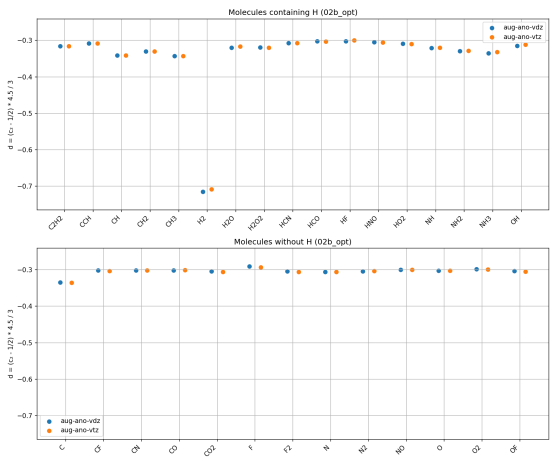


Figure 1: Free floating depths for AE calculations

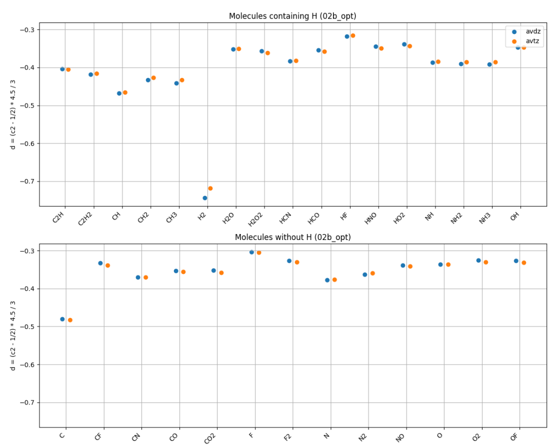


Figure 2: Free floating depths for ECP calculations

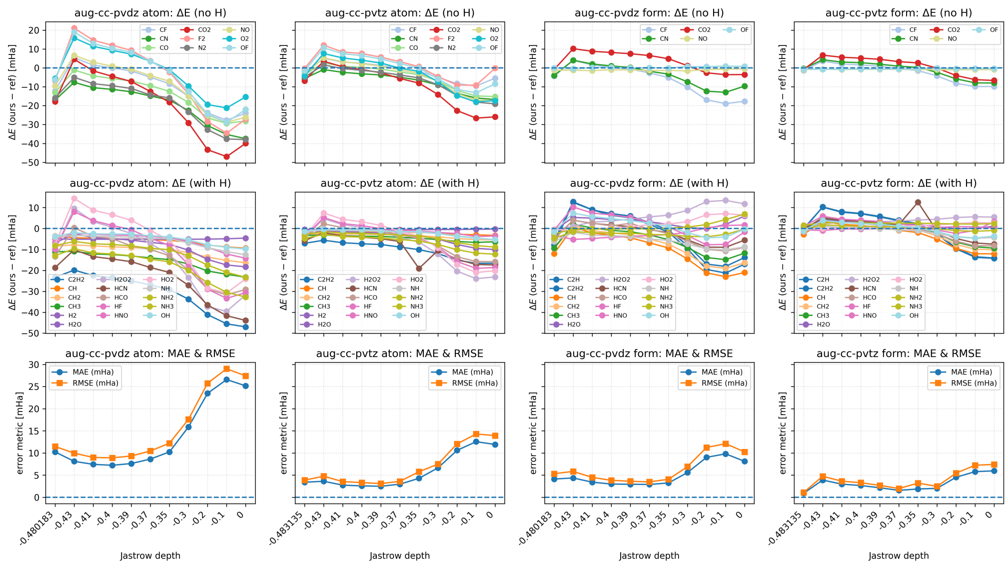


Figure 3: eCEPP

## 0.2 CCSD(T) energies against heat

### 0.2.1 eCEPP

### 0.2.2 ccECP

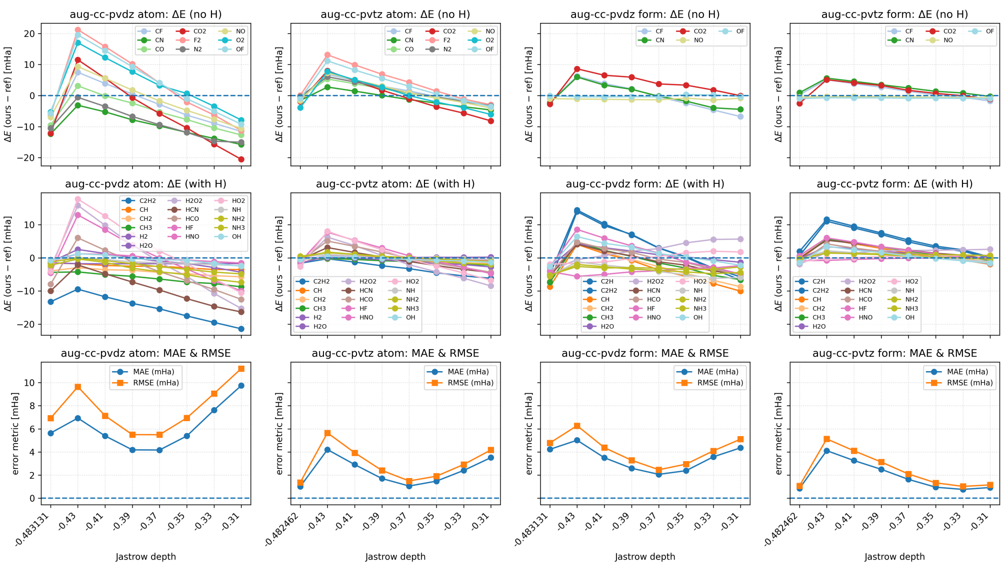


Figure 4: cECP

## 0.3 Depth patterns

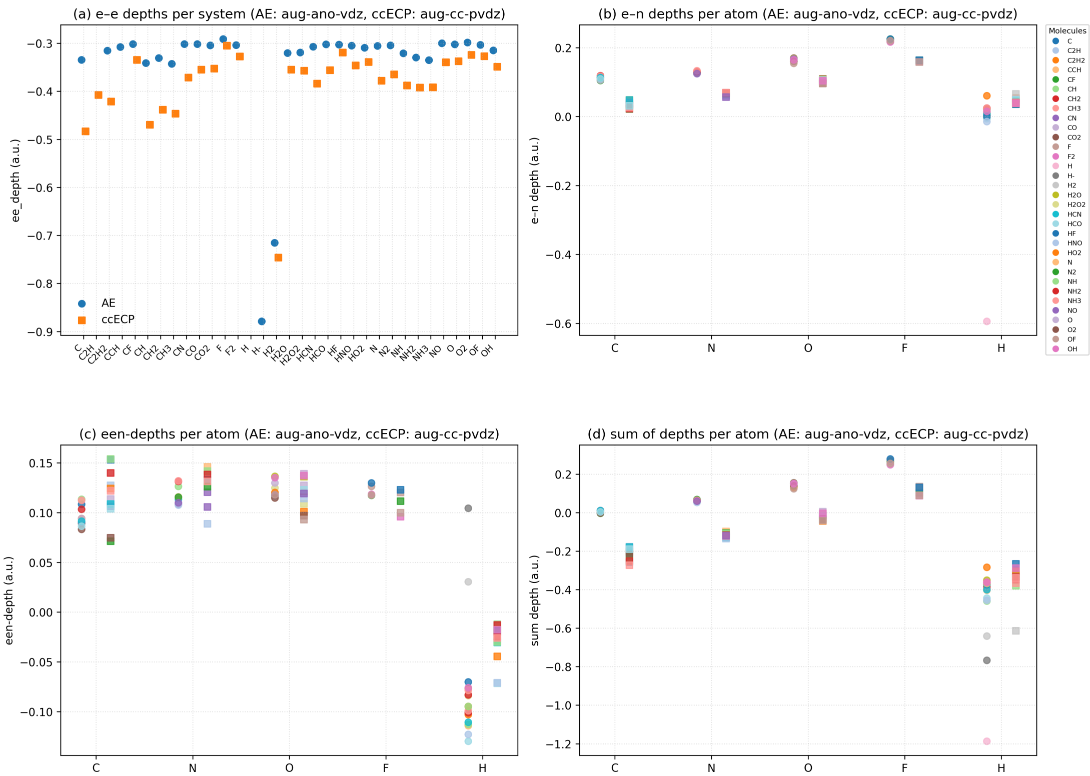


Figure 5: Patterns of free floating depths from varmin optimizations

## 0.4 Cutoff and order sweep analysis

We analyze the CCSD(T) versus HEAT errors (in mHa) from `data/energies_vs_order_cutoff.csv`. The analyzed systems are the molecules listed in the `species` column of that file (at the moment, $CH_3$, $F_2$, HF, $NH_2$, $NH_3$). We vary the electron–nucleus and electron–electron–nucleus cutoffs (en_cutoff, een_cutoff) and the Jastrow orders (ee_order, en_order, een_order), and assess parameter choices by the distribution of per-system errors and by robustness across systems.

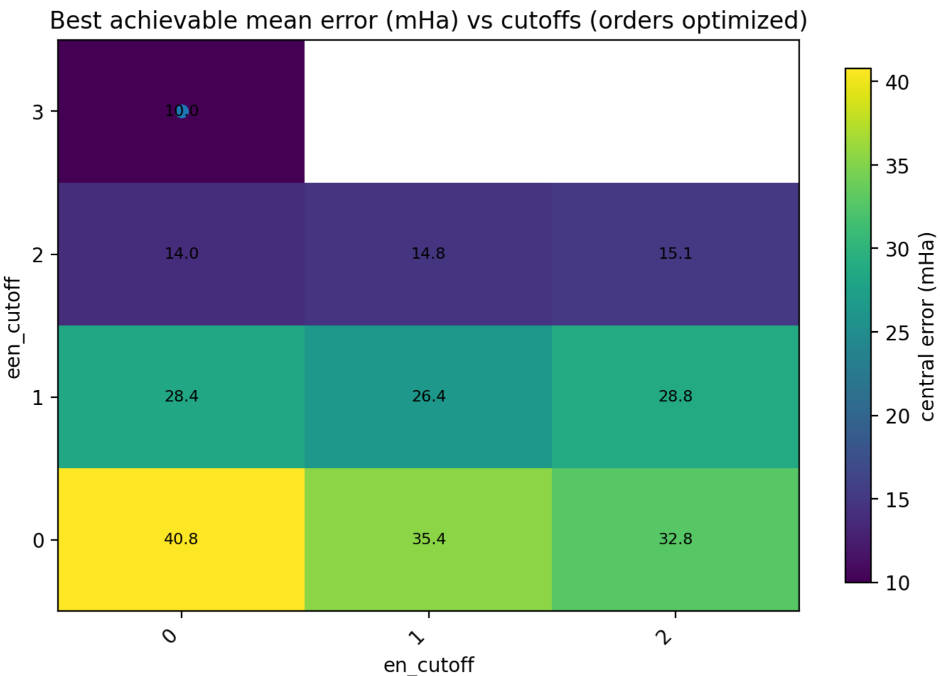


Figure 6: Best achievable aggregate error versus (en_cutoff, een_cutoff) when the Jastrow orders are varied for each cutoff pair. Cell values report the aggregate error in mHa.

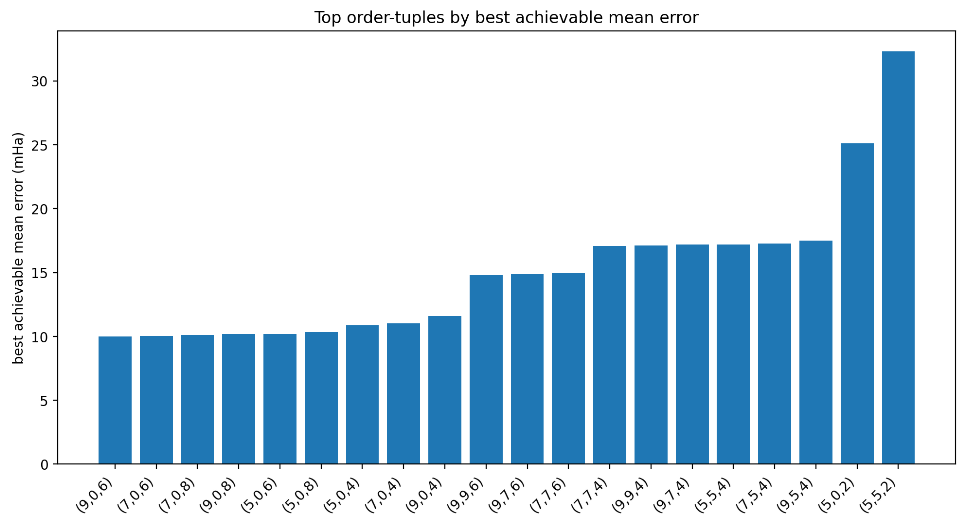


Figure 7: Best achievable aggregate error for each order triple (ee_order, en_order, een_order) with the tested cutoffs. Lower bars indicate better-performing order choices.

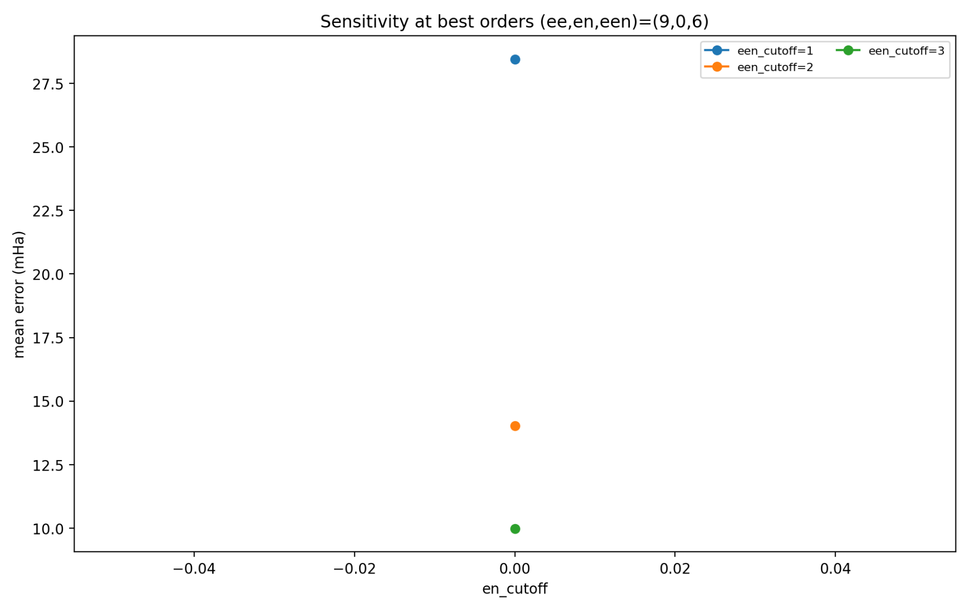


Figure 8: Sensitivity of the aggregate error to en_cutoff at the selected best order triple. Each curve corresponds to a fixed een_cutoff.

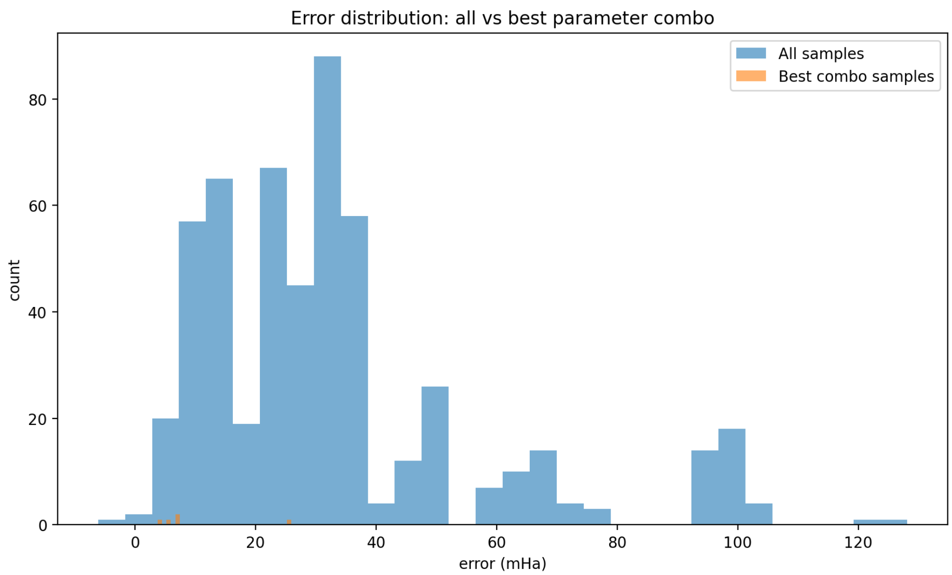


Figure 9: Distribution of individual errors (all rows in the sweep, i.e., each count denotes one molecular calculation and its error) compared to the distribution obtained at the selected best parameter combination (en cutoff=0, een-cutoff = 3, params=(9,0,6)).

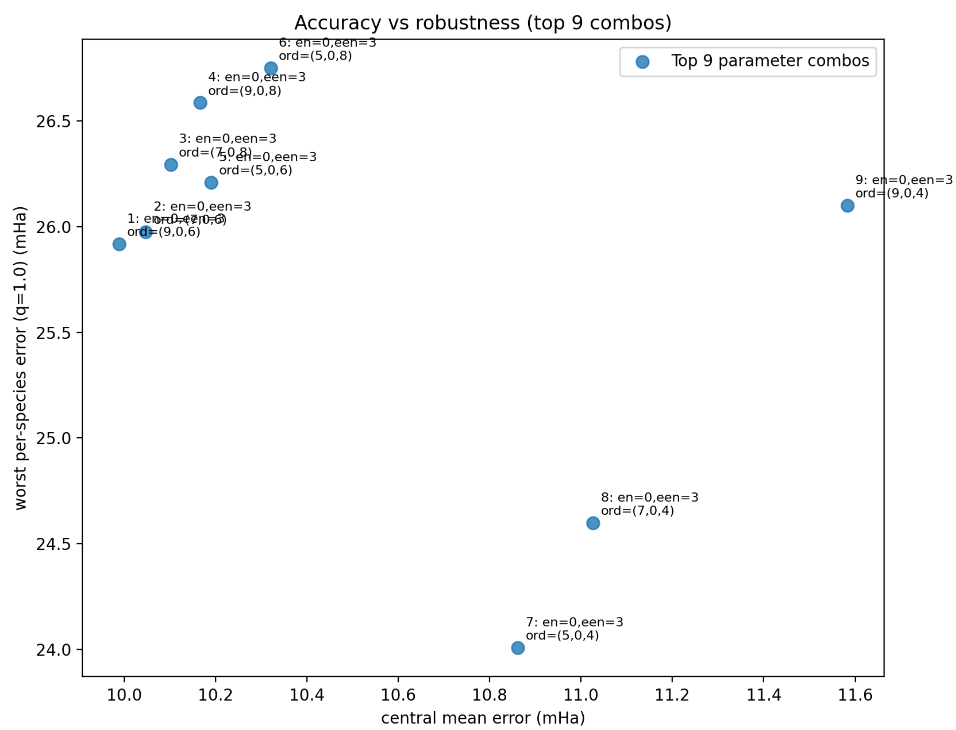


Figure 10: Accuracy versus robustness among the top-ranked parameter combinations. Each labeled point corresponds to one (en_cutoff, een_cutoff, ee_order, en_order, een_order) choice, with coordinates given by the aggregate error and the worst per-system error.

# 1 Introduction